Smaller Diophantine forms of ZFC consistency

Combining blank-input compilation, ordinary Pell norms,
root elimination, and a five-cell local language

Research contribution prepared for Vladimir Reshetnikov's ProveIt repository

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Abstract

We combine and simplify the eight fixed-instance constructions collected in ProveIt's *Con(ZFC) in Diophantine form* report. The principal transparent construction is a system of six polynomial equations in fifteen positive unknowns, of maximum degree 28, with a 60-operation arithmetic evaluator. An auxiliary-gap substitution lowers the maximum degree to 22 at a cost of one additional operation; its sum of squares has degree 44 and costs 78 operations. Combining this gap with Pell-root elimination gives one degree-219 equation in fourteen positive unknowns at 75 operations. A 73-operation, degree-342 alternative offers another tradeoff. A quadratic presentation uses 32 positive unknowns and 23 equations, yielding one quartic.

A separate simplification removes the four unused corners of the local computation windows. The selector alphabet drops from approximately $1.42 \cdot 10^{21}$ to exactly 38 256 607 897, and the bit length of the base coefficient drops from 5^{67} to 5^{36}. All four coefficients are fixed by a terminating finite recipe from the literal K215 transition table. They are not free parameters and are not expanded.

For each presented system or polynomial, ordinary mathematics formalizable in ZFC proves that it has a solution in its stated domain if and only if ZFC has a finite refutation. Thus ZFC proves its unsolvability equivalent to Con(ZFC). The proof includes a sign-independent auxiliary Pell argument and a main-root sign argument; these reduce several unrestricted-integer conversions substantially. We give exact circuits, quadratic residuals, source hashes, reproducible finite audits, and an explicit account of the inherited proof dependencies. No global optimality, consistency proof, or new Lean/Rocq verification is claimed.

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1 What is improved, and what "simpler" means

The requested object is a fixed polynomial or finite system, with integer coefficients, for which

$$\mathrm{ZFC} \vdash [\text{no solution in the stated domain}] \iff \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

There are several incompatible notions of simplicity: number of unknowns, number of equations, degree, arithmetic evaluation length, and coefficient height. We give a small family of concrete improvements and do not identify one of these objectives with all the others.

An arithmetic operation means one binary addition, subtraction, or multiplication, including an operation with a fixed numeral. Reading a port, reusing a computed value, and comparing two values in a system cost zero. Every source packet gives the complete acyclic circuit under this convention. Coefficients are fixed numerals; the table does not charge their bit lengths. The separate coefficient analysis does.

The following primary presentations use strictly positive unknowns. Their coefficients are the *same four new integers* from Section 10; using the old package-06 coefficients instead also preserves the arithmetic conclusions.

Presentation	Unknowns	Equations	Degree	Operations
Gap root norm Φ_{14}^{gap}	14	1	219	75
Root outer norm Ψ_{14}	14	1	342	73
Ordinary system $\mathcal{S}_{60}$	15	6	28 maximum	60
Gap system	15	6	22 maximum	61
Product P_{69}	15	1	99	69
Gap product	15	1	93	70
Sum of squares S_{77}	15	1	56	77
Gap sum of squares	15	1	44	78
Quadratic system $\mathcal{Q}_{32}$	32	23	2 maximum	not optimized
Sum of squares of $\mathcal{Q}_{32}$	32	1	4	not optimized

The fourteen-variable equations are displayed and measured in Sections 5 and 7. The integer-domain presentations appear in Section 11. In particular, there is a polynomial in 32 unrestricted integer unknowns of degree 434, one in 33 of degree 88, and one in 50 of degree 8.

1.1 Comparison of the eight source packages

The repository reconciliation records eight independently prepared packages, all based on the earlier snapshot `cdd582bec13f908df3711fa9e05d1ddac368a5db`. We inspected the reconciled snapshot `3b201cdea32d2c816f5f3d20734051f25fd3ef14`. The following table selects the source constructions that affect the present comparison; it is not a claim that every historical check or formal build was rerun here. Full provenance and the source paths are recorded in the archive and in the reconciliation note [1].

Source	Relevant construction	Role here
01-explicit82	18 variables, one degree-187 equation, 82 operations; fixed-input specialization of the full compiler	Baseline specialization
02-system63	16 variables, seven equations, maximum degree 28, 63 operations; ordinary strong norm	Low-degree Pell block
03-system64	A closely related 64-operation system; elimination of the strong root gives 15 variables	Root-elimination method
04-poly73	16 variables, degree 156, 73 operations; direct two-sided normalizer and affine input guard	Alternative normalization
05-fixed15-16	15-variable norm products: 82 operations/degree 295 and 80 operations/degree 346	Two root finalizers
06-blank67	Raw K215; 15 variables, six equations/61 operations; polynomial 67 operations/degree 155	Blank semantics and source circuit
07-fixed70	16-variable product, 70 operations/degree 169; 17-variable system, 61 operations/degree 28	Proof-order and scale checks
08-start-only67	Same 67-operation arithmetic skeleton with different coefficients; quartic in 34 positive variables	Quadratic lifting

The gains combine compatible interfaces, rather than numerical coefficients from different machines. Package 02's ordinary strong norm can be used after package 06 deletes the entire input interface. Package 08's lifting then loses two registers. Packages 03 and 05's root elimination can be applied *after* deletion of the input interface, giving fourteen unknowns. Finally, the local predicate in package 06 does not inspect its four corner entries. Removing those entries reduces coefficient height without adding any arithmetic gate. The additional substitution $y = V + e$ cancels the highest auxiliary square terms. Applying that substitution before root elimination gives the degree-219 fourteen-variable equation, at the same 75 operations as its degree-293 predecessor.

The 60-operation system improves the earlier 16-variable, seven-equation, 63-operation system at the same maximum degree 28. It also improves the 61-operation, 15-variable blank system in both operations and degree. The quartic improves 34 positive unknowns and 25 quadratic equations to 32 and 23. The 67-operation source remains useful when operation count alone is prioritized; our 69-operation product trades two operations for a degree decrease from 155 to 99.

Classical low-variable universal constructions are a different comparison. Jones's work includes nine-variable universal representations with a very different degree and coefficient tradeoff [13]. The repository's `Jones1982.nine_unknowns` states an existence theorem, not a literal nine-variable specialization of these four ports. Modern relation-combining constructions also give representations in eleven unrestricted integer unknowns, with much larger degrees [14]. We make no absolute minimum-variable or discovery-priority claim.

2 The six equations in fifteen unknowns

Fix the four integers

$$\beta = B - 1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F^{\mathrm{src}} = M_F + B - 1 \tag{2}$$

from Section 10. Here M_F is the native mask; M_F^{src} is the shifted source port. They must not be confused.

The fifteen supplied unknowns are

$$(J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma) \in \mathbb{Z}_{>0}^{15}.$$

Define the following *polynomial abbreviations*. They are not additional existential unknowns:

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & D &= X + ac + \gamma H. \end{aligned} \tag{3}$$

Further set

$$\begin{aligned} G &= (q-1)(q-F) + \alpha = q(q-F) - C, \\ R &= G(q^2-1) + (M_C + qM_F^{\mathrm{src}})J, \\ L &= EkY, \qquad V = c(Tf-1) - Rf^2, \qquad I = ic^2, \qquad U = \Delta(f^2-1). \end{aligned} \tag{4}$$

The system is

$$\boxed{\begin{aligned} \tau^2 - L(L+k) &= 1, \\ D^2 - \Delta c^2 &= 1, \\ U(V^2 - y^2) + y^2 &= 1, \\ k - hE - R &= 1, \\ (K+w)C &= F + z(q-1), \\ I^2 &= U. \end{aligned}} \tag{5}$$

All powers have fixed integer exponents. Evaluation uses only ordinary integer polynomial arithmetic. The source `arithmetic/ordinary_blank_system60.json` has 60 live gates; all residuals have degree at most 28.

Theorem 2.1. *For the four coefficients in Section 10, ZFC proves*

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ \mathcal{S}_{60}(\mathbf{x}) \quad \longleftrightarrow \quad \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).$$

Thus the absence of a positive solution is provably equivalent in ZFC to Con(ZFC)*. The same assertion holds for the other presentations in this article, in their explicitly stated domains.*

2.1 Two single-polynomial presentations

For the product presentation use the same definitions but call the transport unknown z_0, and put

$$\begin{aligned} N_0 &= \tau^2 - L(L+k), & N_1 &= D^2 - \Delta c^2, \\ N_A &= U(V^2 - y^2) + y^2, & N_k &= k - hE - R, \\ N_t &= (K+w)C + q - F - z_0(q-1), & N_s &= 1 + I^2 - U. \end{aligned}$$

Then

$$\boxed{P_{69} = N_0 N_1 N_A N_k N_t N_s - 1 = 0.} \tag{6}$$

Its complete evaluator has 69 operations and exact degree 99. The product and the six-equation system correspond by $z_0 = z + 1$. The inverse shift preserves positivity on solutions; this is proved below and is not assumed on arbitrary tuples.

Alternatively, let $r_1, \ldots, r_6$ be the six left-minus-right residuals in (5). Then

$$\boxed{S_{77} = r_1^2 + \cdots + r_6^2 = 0} \tag{7}$$

has exactly the same zero set over the integers in the stipulated domain. The six residual subtractions, six squarings and five additions add 17 operations to the 60-operation comparison circuit. Its exact degree is 56.

3 Pell facts and the signed auxiliary argument

For $A \geq 2$ define

$$\chi_A(n) + \psi_A(n)\sqrt{A^2-1} = (A + \sqrt{A^2-1})^n \quad (n \geq 0).$$

Both sequences have integer coefficients and obey $u_{n+2} = 2Au_{n+1} - u_n$. Their initial pairs are $(\chi_0, \chi_1) = (1, A)$ and $(\psi_0, \psi_1) = (0, 1)$. Positive norm-one solutions are exactly these pairs. The usual proof divides by the least positive unit until the quotient lies between 1 and that unit. This takes place in the integer-coefficient quadratic ring and needs no assertion about the full ring of algebraic integers.

3.1 The ordinary strong norm really does recover integrality

The equation $(ic^2)^2 = \Delta(f^2-1)$ cannot simply be read as an integral Pell equation at A: divisibility by Δ must be proved. We use the following precise inherited lemma [6], giving its argument to make the dependency explicit.

Lemma 3.1 (Ordinary strong rank)**.** *Let $\Delta = A^2 - 1$, $c = \psi_A(p) > A\Delta^2$, $p \geq 1$, and $i > 0$. If $(ic^2)^2 = \Delta(f^2-1)$ for an integer f, then for an integer $m > 0$,*

$$|f| = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad ic^2 = \Delta\psi_A(m), \quad c^2 \mid f^2 - 1.$$

Proof. Write $\Delta = d_0 s_0^2$ with $d_0 > 1$ squarefree. Let $F + y_0\sqrt{d_0}$ be the least positive integer-coefficient unit of norm one. For some $e \geq 1$,

$$A = \chi_F(e), \quad s_0 = y_0 S, \quad S = \psi_F(e), \quad \Delta = (F^2-1)S^2, \quad \psi_F(ep) = Sc.$$

Put $W = ic^2$. Divisibility in $W^2 = \Delta(f^2-1)$ first gives $s_0 \mid W$, then $d_0 \mid W/s_0$. Thus $|f| + (W/(d_0 s_0))\sqrt{d_0}$ is an integral unit, so for some $n > 0$,

$$|f| = \chi_F(n), \qquad W = (F^2-1)S\psi_F(n).$$

The Pell sequence is strongly divisible: $\gcd(\psi_F(u), \psi_F(v)) = \psi_F(\gcd(u, v))$. This follows from the addition identity, coprimality of the two coordinates of a norm-one unit, and the Euclidean algorithm.

Let $g = \gcd(n, ep)$, $t = \psi_F(g)$, and $T_0 = (F^2-1)S \leq \Delta$. Since $c^2 \mid T_0\psi_F(n)$,

$$c \leq \gcd(c^2, \Delta c) \mid \gcd(T_0\psi_F(n), T_0\psi_F(ep)) = T_0 t,$$

so $t \geq c/\Delta$. If $g < ep$, then $2g \leq ep$. The duplication identity gives $2t^2 < \psi_F(2g) \leq Sc < Ac$, since $S < A$. But $t^2 \geq c^2/\Delta^2 > Ac$, a contradiction. Therefore $ep \mid n$. Writing $n = em$ gives $p \mid m$ and $W = \Delta\psi_A(m)$.

Write $m = pu$. Expansion of $(\chi_A(p) + c\sqrt{\Delta})^u$ modulo c^2 gives

$$\psi_A(pu)/c \equiv u\chi_A(p)^{u-1} \pmod{c}.$$

Thus $c \mid \Delta u$. Also $\gcd(c, \Delta) = \gcd(p, \Delta)$, by the Pell recurrence modulo Δ. This gcd divides p, whence $c \mid pu = m$. Finally $f^2 - 1 = \Delta\psi_A(m)^2 = ic^2\psi_A(m)$, proving the last divisibility. □

3.2 A step-down lemma with all signs visible

Lemma 3.2 (Strict chi step-down)**.** *If $A \geq 2$, $n \geq 0$, $0 < r < m$, and $\chi_A(n) \equiv \chi_A(r) \pmod{\chi_A(m)}$, then $n \equiv r$ or $-r \pmod{2m}$.*

Proof. Modulo $\chi_A(m)$, the addition identities give $\chi_A(n+2m) \equiv -\chi_A(n)$ and $\chi_A(2m-n) \equiv -\chi_A(n)$. Reducing and reflecting the index gives a residue of the form $\pm\chi_A(t)$ with $0 \leq t \leq m$. The case $t = m$ is zero and cannot work. For $t, r < m$, both values are positive and less than $\chi_A(m)/2$, since $\chi_A(m) > 2\chi_A(m-1)$. Their sum is smaller than the modulus, so the negative sign is impossible. Equality of the positive residues gives $t = r$ by strict growth. Undoing the reductions proves the assertion. □

Lemma 3.3 (Signed auxiliary index)**.** *Suppose $c = \psi_A(p)$, $|f| = \chi_A(m)$, $c \mid m$, $m > 2p$, and $0 < p, R < c/2$. Let $I > 1$ be divisible by c, with $I^2 = \Delta(f^2 - 1)$. If integers V, y satisfy*

$$I^2V^2 - (I^2-1)y^2 = 1, \quad V \equiv -c \pmod{|f|}, \quad V \equiv -R \pmod{c},$$

then $p = R$. No sign condition on V or y is needed.

Proof. Neither V nor y can vanish. Pell classification gives $I|V| = \chi_I(\ell)$, $|y| = \psi_I(\ell)$. Modulo I, even-index chi values are ± 1, so $\ell = 2v+1$ is odd. There is an integer polynomial Q_v with

$$\chi_t(2v+1) = tQ_v(t^2), \quad Q_v(1-A^2) = (-1)^v\psi_A(2v+1), \quad Q_v(0) = (-1)^v(2v+1).$$

These identities follow from the second-order recurrence (equivalently, the odd Chebyshev identity). Let $\sigma = \operatorname{sign} V$; then $|V| = \sigma V = Q_v(I^2)$. As $I^2 \equiv 1 - A^2 \pmod{|f|}$, the first congruence gives $\psi_A(\ell)^2 \equiv c^2 \pmod{|f|}$. Using $\chi_A(2t) = 1 + 2\Delta\psi_A(t)^2$, Lemma 3.2 gives $\ell \equiv \pm p \pmod{m}$, hence modulo c. Modulo c, $I^2 \equiv 0$ and the second congruence gives $\sigma(-1)^v\ell \equiv -R \pmod{c}$. Thus $R \equiv \pm p \pmod{c}$. The strict bounds exclude both a nonzero wrap and the negative alternative, leaving $R = p$. □

The lemma explains why one should not require the deleted auxiliary quotients to be positive before extracting the index. It also prevents an invalid use of a positive-root parity lemma after signs have been released. We recover oddness from the first/main index relation; any stronger residue class is recovered later from the actual masks.

4 Proof of the arithmetic reduction

We prove soundness on the larger domain in which only

$$J, F, \alpha, z, i, s, w, \eta, \zeta > 0 \tag{8}$$

are required positive. The six remaining coordinates h, f, T, τ, y, γ may be arbitrary integers. For the product, replace z by the positive z_0. This stronger theorem supplies the integer conversions without inserting unnecessary positivity guards.

4.1 Units, the boundary case, and pre-decoding bounds

For the system, transport immediately implies $C > 0$. For the product, each of its six integer factors is ± 1. The first factor cannot be -1: the negative Pell equation for $u(u+1)$, $u = XY^2 > 1$, has no solution. A least positive second coordinate would be decreased by multiplication by $(2u+1) - 2\sqrt{u(u+1)}$.

Since $\Delta = (a+2)^2 - 1$ is 0 or 3 modulo 4, the main norm cannot be -1. Nor can $N_s = 1 + I^2 - \Delta(f^2-1)$: modulo 4 it is either $1 + I^2$ or $I^2 + f^2$, neither of which is 3. Thus $N_s = 1$, giving $U = I^2 > 0$. The auxiliary factor is now congruent to either V^2 or y^2 modulo 4, and also equals 1. Only $N_k = \varepsilon$ and $N_t = \nu$ remain, with $\varepsilon, \nu \in \{1, -1\}$ and $\varepsilon\nu = 1$.

Weak transport gives $C \geq 0$, since

$$q - F - z_0(q-1) = 1 - F - (z_0 - 1)(q-1) \leq 0$$

and $C \leq -1$ would force $N_t < -1$. We retain $C = 0$ at this stage. It really is an algebraically possible boundary for the negative transport unit and must not be discarded by assertion.

In both presentations, $q \geq B \geq 16$, $F \geq 1$, $\alpha \geq 1$, and $q - F = C + \alpha > 0$. Therefore

$$q \leq G \leq q^2 - q, \qquad 0 < (M_C + qM_F^{\mathrm{src}})J < 2q^2 - q - 1.$$

The native mask bounds used here are $0 < M_C, M_F < B - 1$, with $M_F^{\mathrm{src}} = M_F + B - 1$. Consequently

$$q(q^2-1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \qquad E \geq q^4 > R + 2, \qquad a > R + 2. \tag{9}$$

These bounds precede every dyadic, Pell-index, or computation-decoding conclusion.

4.2 Bootstrap, exact index, and the two remaining signs

Put $A = a + 2$ and $P_0 = 2XY^2 + 1$. The first norm gives

$$|\tau| = \chi_{P_0}(n), \qquad k = 2\psi_{P_0}(n), \qquad n \geq 1.$$

Since $P_0 \equiv 1 \pmod{E}$, the index unit gives

$$2n \equiv R + \varepsilon \pmod{E}, \quad 2n = R + \varepsilon + vE \quad (v \geq 0), \quad n \geq (R-1)/2 \geq 24.$$

This does not require a sign for h. The main norm gives $|D| = \chi_A(p)$, $c = \psi_A(p)$. Since $P_0 > A$ and $c > k$, monotonicity gives $p > n$. Elementary growth now implies

$$c > A\Delta^2, \qquad c > 2p, \qquad c > 2R.$$

For the last inequality one may already use $p \geq 2$ and $c \geq 2A > 2R$; no positivity of h is hidden in it.

Lemma 3.1 applies to $I = ic^2$, yielding the auxiliary index m, its divisibility, and $c^2 \mid f^2 - 1$. The formula for V, even when T and f are signed, gives

$$V \equiv -c \pmod{|f|}, \qquad V \equiv -R \pmod{c}.$$

Lemma 3.3 therefore gives $p = R$.

If $v \geq 1$, then $2n \geq R - 1 + E > 2R$, contrary to $n < p = R$. Thus $2n = R + \varepsilon$. If $\varepsilon = -1$, then $p = 2n + 1$. Let $Q = \chi_A(2) = 2A^2 - 1 > P_0$. Duplication gives

$$c = \psi_A(2n+1) > \psi_A(2n) = 2A\psi_Q(n) \geq 2A\psi_{P_0}(n) = Ak > k(Y+1),$$

contrary to $kY < c < k(Y+1)$. Hence $\varepsilon = \nu = 1$. In the product presentation this now excludes $C = 0$, so $C > 0$. We have $R = 2n - 1$, an odd positive integer.

4.3 The main root sign and the power of two

The sign of D matters if γ is unrestricted. Write $R = 2r + 1$, $r = n - 1$. The actual lower bound in (9) gives $r \geq 24$. Standard recurrence bounds give

$$\psi_b(t) \geq (2b-1)^{t-1}, \qquad \psi_b(t) \leq (2b)^{t-1} \quad (b \geq 2, t \geq 1).$$

Using $(2A-1)^2 > 2P_0X$, the strict ratio implies

$$Y + 1 > \frac{c}{k} = \frac{\psi_A(2r+1)}{2\psi_{P_0}(r+1)} > \frac{X^r}{2}.$$

Thus

$$Y > X^r/3 > 2^R, \qquad Y > X, \qquad a > 2^R. \tag{10}$$

The inequalities use only $X \geq 16$ and $r \geq 24$.

Modulo the odd integer $H = 4a + 3$, the two projected Pell sequences obey

$$\chi_A(t) - a\psi_A(t) \equiv 2^t, \qquad \chi_A(t) + a\psi_A(t) \equiv 2^{-t}.$$

One checks the initial values and recurrence; $2(a+1)$ is the inverse of 2 modulo H. If $D < 0$, its defining equation would give $X2^R \equiv -1 \pmod{H}$. But $0 < X2^R < XY = E < a < H - 1$, a contradiction. Therefore $D > 0$. The first projected congruence now gives $X \equiv 2^R \pmod{H}$. Both are between 0 and $a < H$, hence

$$\boxed{X = 2^R.} \tag{11}$$

In particular q is a power of two, and $(B-1) \mid (q-1)$ implies $q = B^N$ for an integer $N \geq 1$.

Also $h > 0$ and $\gamma > 0$ are consequences, not domain assumptions. A negative h contradicts $k > 0$ and $E > R + 2$; $h = 0$ contradicts first-norm growth. Finally

$$D - ac = 2c - \psi_A(R-1) > c > Y > X,$$

so the equation defining D forces $\gamma > 0$.

4.4 The inherited bit-count and native compiler interfaces

At this point the first/main data satisfy the hypotheses of the post-index ratio and valuation part of the half-binomial Pell argument [9, 8]: its scale assumptions are $X = wq$, $Y = sq^3$, $q \geq 16$, and $3q + 1 \leq R < q^4$, with the exact odd index and strict ratio just proved. Its input-independent conclusion is

$$\nu_2(Y) = \mathrm{pc}(R) - 2.$$

Here is the precise tail of the argument being used. Put $\xi = (X+1)^{2r}/X^r$. The inherited ratio estimate gives

$$0 < \frac{c}{k} - \frac{\xi}{2} < \frac{16r}{X+1} < \frac{1}{2}.$$

Expand $\xi = M + \theta$, where

$$M = \binom{2r}{r} + \sum_{j=1}^{r} \binom{2r}{r+j} X^j, \qquad 0 < \theta < \frac{1}{4}.$$

The integer M is even. Thus $Y < c/k < Y + 1$ gives $Y = M/2$. The central-binomial valuation identity $\nu_2\binom{2r}{r} = \mathrm{pc}(r)$, together with $X = 2^{2r+1}$, now gives $\nu_2(Y) = \mathrm{pc}(r) - 1 = \mathrm{pc}(R) - 2$. This uses no auxiliary positivity or positive-root parity assertion. If the symmetric version is used,

$R > 3q$ and $X = 2^R$ imply $q^3 \mid X$, restoring its scale by a positive integer rescaling of w. Since $q^3 \mid Y$, writing $q = 2^t$ gives $\mathrm{pc}(R) \geq 3t + 2$.

The shifted packing identity is

$$\begin{aligned} S' &= C + qF - 1, \\ T' &= (M_C J + 1) + q(M_F J - 1), \\ R &= (q^2 - S')(q^2 - 1) + T'. \end{aligned}$$

Here $0 < S' < q^2$ and the two mask words have the precise native populations. The inverse-population lemma [10, 4] gives

$$(C - 1) \,\&\, (M_C J + 1) = 0, \qquad F \,\&\, (M_F J - 1) = 0.$$

These are decoded consequences, not operations in the polynomial. Transport is the required native congruence because $(K + X)C - F$ differs from $(K + w)C - F$ by a multiple of $q - 1$. Sections 9–10 establish the remaining finite-language and coefficient hypotheses. The masks force a Start-containing valid blank tableau, hence a halting K215 run.

The half-binomial, inverse-population, and canonical actual-index completeness lemmas are inherited mathematical dependencies. Their statements, pinned files, and relevant source proofs are included or identified in `provenance.json`. The finite scripts in this archive are not substitutes for those unbounded arguments.

4.5 Completeness and the positive transport shift

Given a blank halting run, the native compiler's canonical padding and actual-index construction supplies the normalized blank witnesses of package 06 [4]. Keep every coordinate except the ordinary strong multiplier and set $i_{\mathrm{ordinary}} = \Delta i_{\mathrm{normalized}}$. Then

$$(i_{\mathrm{ordinary}} c^2)^2 = \Delta(f^2 - 1),$$

and the auxiliary coefficient has the same value. Thus the ordinary system with the old transport factor has a positive solution.

On this solution, $w = 2^R/q > q > F$, because $R > 3q$. As $C \geq 1$,

$$(z_0 - 1)(q - 1) = (K + w)C - F > 0.$$

Hence $z_0 \geq 2$, and $z = z_0 - 1$ is positive. This is the one-operation transport saving in (5). The reverse shift is immediate. Completeness on the larger signed domain follows because the canonical witnesses are positive.

5 Fourteen unknowns by eliminating the strong root

The blank-input reduction also combines with the root-elimination method of packages 03 and 05. This removes one supplied unknown, at the cost of a higher-degree equation. In this section i has its *normalized* meaning: the root to be eliminated would satisfy $f^2 = 1 + \Delta i^2 c^4$. The ordinary multiplier used in Section 4 will be Δi.

Delete f from the fifteen-variable list and keep all definitions that do not involve it. Use the old positive transport coordinate z_0, so that $N_t = (K + w)C + q - F - z_0(q - 1)$. Define

$$\begin{aligned} Q_* &= (\Delta i c^2)^2, & d_* &= 1 + \Delta i^2 c^4, \\ b_* &= c + R d_*, & v_* &= cT, \\ A_* &= Q_*(d_* v_*^2 + b_*^2 - y^2) + y^2 - 1, \\ B_* &= 2Q_* v_* b_*, & K_* &= A_*^2 - d_* B_*^2, \\ P_4 &= N_0 N_1 N_k N_t. \end{aligned} \tag{12}$$

All these are polynomial abbreviations in fourteen supplied unknowns; none introduces another existential coordinate. The first equation is

$$\boxed{\Phi_{14} = P_4(1 + K_*) - 1 = 0.} \tag{13}$$

Its circuit has 75 operations and exact degree 293.

A second finalizer incorporates the outer product inside the norm. Put

$$A_{\text{out}} = P_4\big[Q_*(d_* v_*^2 + b_*^2) + (1 - Q_*)y^2\big] - 1,$$
$$B_{\text{out}} = P_4 B_*.$$

Then

$$\boxed{\Psi_{14} = A_{\text{out}}^2 - d_* B_{\text{out}}^2 = 0} \tag{14}$$

has 73 operations and exact degree 342. Both equations initially use fourteen strictly positive unknowns. Their coefficients are the same four fixed integers as before.

The circuits compute the norm cross-term through

$$d_* B_*^2 = 4(Q_* d_* v_*^2)(Q_* b_*^2),$$

and its analogous outer-product expression. Thus they do not pay to construct B_* separately. Neither circuit contains division, square-root extraction, variable exponentiation, or a free program index.

5.1 Restoring an integer root, with its sign left free

At a zero of (13), both P_4 and $1 + K_*$ are integer units. Because B_* is even,

$$1 + K_* \equiv 1 + A_*^2 \pmod 4,$$

which cannot be -1. Hence $K_* = 0$ and $P_4 = 1$. Every factor inside P_4 is consequently a unit. The first and main norms equal 1 by the descent and congruence arguments already given. The transport-unit argument yields $C \geq 0$, and the pre-decoding bounds (9) apply; in particular $R > 0$. On the positive domain, $Q_*, v_*, b_* > 0$, so $B_* \neq 0$.

The rational number

$$f = \frac{A_*}{B_*}$$

satisfies $f^2 = d_*$. A rational square root of an integer is an integer: write it in lowest terms, and any prime in its denominator would have negative valuation in its square. Since $d_* > 1$, this restored integer f is nonzero. Its sign need not be determined. Set $V = v_* f - b_*$. Using $f^2 = d_*$, the exact remainder identity is

$$A_* - B_* f = Q_*(V^2 - y^2) + y^2 - 1 = 0. \tag{15}$$

Moreover

$$V = c(Tf - 1) - Rf^2, \qquad f^2 - \Delta i^2 c^4 = 1.$$

Thus an integer solution of the normalized blank equations has been restored, allowing a signed f. Replace the multiplier by $i_{\text{ordinary}} = \Delta i > 0$. Then

$$(i_{\text{ordinary}} c^2)^2 = \Delta(f^2 - 1) = Q_*.$$

All factors of (6) have the required product, so the signed-domain theorem of Section 4 recovers a genuine K215 blank halting run. No separate positivity theorem for the reconstructed root is required.

For (14), first note that $P_4 = 0$ would make its left side equal 1. Otherwise reduction modulo P_4 gives $\Psi_{14} \equiv 1 \pmod{P_4}$, so a zero forces $P_4 = \pm 1$. Its four factors are units and give the same outer bounds. Since $B_{\mathrm{out}} \neq 0$, set $f = A_{\mathrm{out}}/B_{\mathrm{out}}$. Again $f^2 = d_*$ makes f an integer. The corresponding remainder identity is

$$A_{\mathrm{out}} - B_{\mathrm{out}} f = P_4\big[Q_*(V^2 - y^2) + y^2\big] - 1 = 0.$$

After the same change of strong multiplier, this is exactly the product needed by (6). The argument does not assume the sign of P_4 in advance.

Conversely, a canonical positive blank compiler solution has $f^2 = d_*$, $A_* = B_* f$, and $P_4 = 1$. Deleting f therefore makes both (13) and (14) vanish. This proves the two directions of their existence equivalence and hence their stated Con(ZFC) equivalences.

5.2 A signed extension and 32 unrestricted integer unknowns

The nine coordinates

$$J, F, \alpha, z_0, i, s, w, \eta, \zeta$$

can remain positive while the other five, h, T, τ, y, γ, are arbitrary integers. The only new obstruction to the preceding argument would be $T = 0$, making the reconstructed-root denominator vanish. We exclude it before taking a quotient.

For either polynomial the outer unit argument already gives $0 < R < a < \Delta$ and $c > 2$. Since $Q_* = \Delta(d_* - 1)$,

$$\begin{aligned} Q_* - b_* &= (\Delta - R)d_* - \Delta - c \\ &\geq d_* - \Delta - c = 1 + \Delta(i^2c^4 - 1) - c > 0. \end{aligned}$$

Consequently $2 < b_* < Q_*$. If $T = 0$, then $v_* = B_* = 0$. For $\Phi_{14} = 0$, the equation $K_* = 0$ forces $A_* = 0$, and therefore

$$Q_* b_*^2 - (Q_* - 1)y^2 = 1. \tag{16}$$

For $\Psi_{14} = 0$, the equation $A_{\mathrm{out}} = 0$ says that the left side of (16) equals $P_4 = \pm 1$. It cannot equal -1, since Q_* is a square and this expression is a square modulo four. Thus (16) follows again.

Every solution of this equation with $b_* > 1$ has $b_* \geq 2Q_* - 1$. Indeed,

$$b_*^2 - 1 = (Q_* - 1)(y^2 - b_*^2)$$

implies $|y| \geq b_* + 1$, and for $1 < b_* \leq 2Q_* - 2$ one has $b_*^2 - 1 < (Q_* - 1)(2b_* + 1)$, a contradiction. This conflicts with $b_* < Q_*$. Hence $T \neq 0$, the root denominators are nonzero, and the signed reconstruction is valid. The signed ordinary theorem then supplies soundness exactly as above.

Use the positivity polynomial (46) independently for the nine guarded coordinates and leave the other five unrestricted. The resulting polynomials have $9 \cdot 3 + 5 = 32$ integer unknowns. The conversion is surjective onto the required positive values, so it preserves existence in both directions. Their operation counts and degrees are included below. They improve the variable count of the explicit family studied here; they do not assert a global minimum over all Diophantine representations.

5.3 Exact degrees and operation ledgers

Write $Q_0 = \beta J$, $k_0 = \eta + \zeta$, and $T_2 = w(Q_0 - F - \alpha) - z_0 Q_0$. The two leading homogeneous forms are respectively

$$(\Phi_{14})_{293} = 8h\gamma i^{12} k_0^{27} w^{21} s^{48} Q_0^{177} (Q_0 - F)^4 T_2, \tag{17}$$

$$(\Psi_{14})_{342} = 64h^2\gamma^2 i^{12} k_0^{30} w^{26} s^{56} Q_0^{206} (Q_0 - F)^4 T_2^2. \tag{18}$$

They are nonzero for every $\beta > 0$. For example, the coefficients at

$$J^{182}h\gamma i^{12}\eta^{27}w^{21}s^{48}z_0, \qquad J^{212}h^2\gamma^2 i^{12}\eta^{30}w^{26}s^{56}z_0^2$$

are $-8\beta^{182}$ and $64\beta^{212}$, respectively.

To check the degree accounting, Q_*, d_*, b_*, v_* have degrees $46, 34, 38, 6$. The unique top term of A_* is $Q_* b_*^2$, of degree 122. Hence K_* has degree 244: the competing term $d_* B_*^2$ has degree 214. The degree-49 outer product gives $244 + 49 = 293$. In the second construction the squared leading term has degree 342 and its subtracted cross-term degree 312.

For the integer versions assign weight two to the nine guarded coordinates and weight one to the five signed coordinates. The leading forms above have signed weights two and four. Their exact degrees therefore become $2 \cdot 293 - 2 = 584$ and $2 \cdot 342 - 4 = 680$. Direct weighted upper bounds give the same values. Disjoint quadratic positivity substitutions preserve nonvanishing, as explained in Section 11.

Presentation	Domain	Unknowns	Degree	Operations
Φ_{14}	positive	14	293	75 = 44M+31A
Ψ_{14}	positive	14	342	73 = 44M+29A
Φ_{14} after positivity substitution	integers	32	584	138
Ψ_{14} after positivity substitution	integers	32	680	136

The integer conversion adds seven gates for each of nine positivity substitutions. The files `arithmetic/build_fourteen.py` and its emitted packets give complete evaluators and reproduce the counts, root-remainder identities, substitution identities, and degree diagnostics. These finite checks supplement the proofs; they do not replace the unbounded compiler or Pell arguments.

6 Lower degree through an auxiliary gap

A further change of coordinate lowers the degrees of the ordinary construction. Replace the supplied y by a supplied e, and compute $y = V + e$. The auxiliary factor becomes

$$N_A^{\mathrm{gap}} = U(V^2 - (V+e)^2) + (V+e)^2 = V^2 - (U-1)e(2V+e). \tag{19}$$

Keep the other factors and residuals unchanged. Thus the new product is

$$P_{70}^{\mathrm{gap}} = N_0 N_1 N_A^{\mathrm{gap}} N_k N_t N_s - 1 = 0.$$

and the new six-equation system is

$$\boxed{\begin{aligned} N_0 = N_1 = N_A^{\mathrm{gap}} = N_k = 1, \\ (K+w)C = F + z(q-1), \\ I^2 = U. \end{aligned}} \tag{20}$$

It replaces only the third equation of (5) by $N_A^{\mathrm{gap}} = 1$. The sum of its six squared residuals gives a third presentation. The circuit needs just one additional addition: its existing y-square now squares the computed value $V + e$.

The signed-domain equivalence is immediate from the inverse integer substitutions $e = y - V$ and $y = V + e$. The nine coordinates $J, F, \alpha, z, i, s, w, \eta, \zeta$ remain positive, while h, f, T, τ, e, γ may be arbitrary integers. Soundness therefore follows from Section 4. Positive-domain completeness needs one extra check. The canonical positive auxiliary Pell point used by the blank compiler has

$V, y > 0$, $U > 1$, $c > 2R$, and $V \equiv -R \pmod{c}$. Hence $V \geq c - R > 1$, and its norm equation gives

$$(U-1)(y^2 - V^2) = V^2 - 1 > 0.$$

Thus $y > V$ and the canonical gap $e = y - V$ is strictly positive. This proves the same existence equivalence in both stated domains.

Presentation	Unknowns	Operations	Exact degree
Six equations	15 positive	61	maximum 22
Product	15 positive	70	93
Sum of squares	15 positive	78	44
Six equations	33 integer	124	maximum 44
Product	33 integer	133	179
Sum of squares	33 integer	141	88

For exact degree certificates, set $Q_0 = \beta J$, $k_0 = \eta + \zeta$, and $T_2 = w(Q_0 - F - \alpha) - z_0 Q_0$. The auxiliary leader is $-2ek_0w^2s^3Q_0^{11}Tf^3$; the six factor degrees are $22, 18, 22, 7, 2, 22$. Consequently

$$(P_{70}^{\mathrm{gap}})_{93} = -16Q_0^{52}h\gamma i^2k_0^8w^7s^{15}Tf^3eT_2. \tag{21}$$

The coefficient of $J^{53}h\gamma i^2\eta^8w^7s^{15}Tf^3ez_0$ is $16\beta^{53} \neq 0$. The degree-22 leaders of the first, auxiliary and strong residuals are

$$-w^2k_0^2s^4Q_0^{14}, \qquad -2ek_0w^2s^3Q_0^{11}Tf^3, \qquad i^2k_0^4s^4Q_0^{12}.$$

Their squared sum is nonzero, proving degree 44 for the sum of squares.

For the integer conversion, give weight two to the nine positive coordinates and weight one to the other six. The residual weights are $44, 35, 39, 13, 4, 44$. Nine independent substitutions (46) add 63 gates and give 33 integer unknowns. The product has degree 179, while the first and strong squared residuals give degree 88 for the sum of squares. The disjoint homogeneous quadratic substitutions preserve the nonzero leading forms. Complete circuits, ledgers and checks are emitted by `arithmetic/build_gap.py`.

7 The degree-219 equation in fourteen unknowns

The auxiliary gap can be introduced *before* eliminating the normalized strong root. This improves the degree-293 inner norm in Section 5 to degree 219 at the same 75-operation cost. It still uses fourteen positive unknowns. Keep normalized i, delete f, replace y with e, and define

$$\begin{aligned}
Q_* &= (\Delta i c^2)^2, & d_* &= 1 + \Delta i^2 c^4,\\
b_* &= c + Rd_*, & v_* &= cT,\\
S_* &= Q_* - 1, & r_* &= b_* + S_* e,\\
p_* &= d_* v_*^2, & B_*^{\mathrm{gap}} &= 2v_* r_*,\\
A_*^{\mathrm{gap}} &= p_* + r_*^2 - Q_* S_* e^2 - 1, & &\\
K_*^{\mathrm{gap}} &= (A_*^{\mathrm{gap}})^2 - d_* (B_*^{\mathrm{gap}})^2, & P_4 &= N_0 N_1 N_k N_t.
\end{aligned} \tag{22}$$

Here the four factors use the old transport coordinate z_0. The proposed equation is

$$\boxed{\Phi_{14}^{\mathrm{gap}} = P_4(1 + K_*^{\mathrm{gap}}) - 1 = 0.} \tag{23}$$

Its evaluator computes the cross-term as $4p_*r_*^2$, so B_*^{gap} is a proof abbreviation and needs no separate register. Removing f and the old y-square leaves 49 prefix operations; the new tail costs 26. The ledger is 43 multiplications and 32 additions or subtractions.

The degree improvement comes from the exact cancellation

$$A_*^{\mathrm{gap}} = p_* + b_*^2 + (Q_* - 1)e(2b_* - e) - 1. \tag{24}$$

If $f^2 = d_*$ and $V = v_*f - b_*$, then

$$A_*^{\mathrm{gap}} - B_*^{\mathrm{gap}} f = Q_*\left[V^2 - (V+e)^2\right] + (V+e)^2 - 1. \tag{25}$$

Thus the new norm eliminates precisely the gap auxiliary equation.

7.1 Restoration and the signed-domain denominators

At a zero of (23), both P_4 and $1 + K_*^{\mathrm{gap}}$ are integer units. Because B_*^{gap} is even, the latter is congruent to $1 + (A_*^{\mathrm{gap}})^2$ modulo four and cannot be -1. Hence $P_4 = 1$ and $K_*^{\mathrm{gap}} = 0$. The outer unit arguments already give $N_0 = N_1 = 1$ and $0 < R < a < \Delta$. On the positive domain, $v_*, b_*, r_* > 0$, so the denominator in $f = A_*^{\mathrm{gap}}/B_*^{\mathrm{gap}}$ is nonzero. This rational root satisfies $f^2 = d_*$, hence is an integer. Identity (25), with $y = V + e$, restores the normalized auxiliary equation. Replacing the multiplier by $i_{\mathrm{ordinary}} = \Delta i$ gives

$$(i_{\mathrm{ordinary}}c^2)^2 = \Delta(f^2 - 1) = Q_*.$$

The signed ordinary theorem therefore proves soundness without imposing a sign on the recovered f or y.

For the expanded domain, keep only $J, F, \alpha, z_0, i, s, w, \eta, \zeta > 0$, and allow $h, T, \tau, e, \gamma \in \mathbb{Z}$. We must now exclude two possible zero factors in $B_*^{\mathrm{gap}} = 2v_*r_*$. The stronger version of the earlier outer bound is

$$\begin{aligned} Q_* - b_* &= (\Delta - R)d_* - \Delta - c \\ &\geq 1 + \Delta(i^2c^4 - 1) - c > 1. \end{aligned}$$

Indeed $c > 2$, $i \geq 1$, and $\Delta > 0$ are already known. Thus $2 < b_* < Q_* - 1 = S_*$, so

$$r_* = b_* + S_*e \neq 0 \qquad \text{for every integer } e.$$

If $T = 0$, then $v_* = p_* = B_*^{\mathrm{gap}} = 0$, and $K_*^{\mathrm{gap}} = 0$ forces $A_*^{\mathrm{gap}} = 0$. Equivalently,

$$Q_*b_*^2 - (Q_* - 1)(e - b_*)^2 = 1. \tag{26}$$

Writing $y_0 = e - b_*$, the inequality $b_* > 1$ would imply $|y_0| \geq b_* + 1$, and hence

$$b_*^2 - 1 = (Q_* - 1)(y_0^2 - b_*^2) \geq (Q_* - 1)(2b_* + 1).$$

This contradicts $1 < b_* < Q_*$. Therefore $T \neq 0$, both denominator factors are nonzero, and the signed reconstruction is valid.

For completeness, the canonical positive blank witness satisfies $f^2 = d_*$, $P_4 = 1$, and $e = y - V > 0$, as proved for the gap auxiliary coordinate. Equation (25) then gives $A_*^{\mathrm{gap}} = B_*^{\mathrm{gap}} f$, so deleting f yields a positive zero of (23). This proves both directions of the same blank-halting and Con(ZFC) equivalence.

Nine independent positivity substitutions (46), with the other five coordinates left unrestricted, give a polynomial in $9 \cdot 3 + 5 = 32$ integer unknowns and 138 arithmetic operations.

7.2 Exact degrees and the outer-norm comparison

For the degree calculation, the following table also records weights two on the nine positive coordinates and one on the five signed ones.

Quantity	Q_*	d_*	b_*	v_*	p_*	r_*	A_*^{gap}	B_*^{gap}	P_4
Degree	46	34	38	6	46	47	85	53	49
Weight	92	68	76	11	90	93	169	104	96

The unique highest term of (24) is $2Q_*eb_*$. Its square has degree 170 and weight 338, while $d_*(B_*^{\mathrm{gap}})^2$ has degree 140 and weight 276. Consequently the new equation has exact degree $49 + 170 = 219$, and its integer conversion has exact degree $96 + 338 = 434$.

Using $Q_0 = \beta J$, $k_0 = \eta + \zeta$, and $T_2 = w(Q_0 - F - \alpha) - z_0 Q_0$, the leading form is

$$(\Phi_{14}^{\mathrm{gap}})_{219} = 32h\gamma e^2 i^8 k_0^{19} w^{17} s^{36} Q_0^{131} (Q_0 - F)^2 T_2. \tag{27}$$

Its coefficient at $J^{134}h\gamma e^2 i^8 \eta^{19} w^{17} s^{36} z_0$ is $-32\beta^{134} \neq 0$. This is also the weighted leading form before the homogeneous positivity substitution, which preserves nonvanishing.

For comparison, the outer-norm variant is

$$A_{\mathrm{out}}^{\mathrm{gap}} = P_4(A_*^{\mathrm{gap}} + 1) - 1, \qquad B_{\mathrm{out}}^{\mathrm{gap}} = P_4 B_*^{\mathrm{gap}},$$

$$\Psi_{14}^{\mathrm{gap}} = (A_{\mathrm{out}}^{\mathrm{gap}})^2 - d_*(B_{\mathrm{out}}^{\mathrm{gap}})^2 = 0.$$

It also costs 75 operations, but has degree 268; its 32-integer-variable conversion has 138 operations and degree 530. The leading form is

$$256h^2\gamma^2 e^2 i^8 k_0^{22} w^{22} s^{44} Q_0^{160} (Q_0 - F)^2 T_2^2.$$

This valid form is dominated by (23) at the same gate count. It does not dominate the earlier 73-operation outer norm.

Its soundness proof is included to make that comparison independently checkable. Reduction modulo P_4 forces $P_4 = \pm 1$, giving the same outer bounds and $r_* \neq 0$. If $T = 0$, its equation says that P_4 times the left side of (26) equals one. That bracket cannot be -1 modulo four because Q_* is a square; therefore (26) holds, an impossibility. Its root denominator is consequently nonzero. The integer restored root now satisfies

$$A_{\mathrm{out}}^{\mathrm{gap}} - B_{\mathrm{out}}^{\mathrm{gap}} f = P_4\{Q_*[V^2 - (V + e)^2] + (V + e)^2\} - 1 = 0.$$

This restores the full normalized product, hence the signed ordinary one. The canonical witness proves completeness as before.

Both gap root variants, their integer conversions, their live-gate ledgers, and their algebraic checks are emitted by `arithmetic/build_gap.py`. The separate proof note records the complete coordinate substitutions and nonzero-denominator arguments.

8 A quadratic system in thirty-two unknowns

Retain the same fixed coefficients. Use the 32 positive unknowns

$$J, F, \alpha, z, f, h, i, s, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, G, L, F_2, V, g, I, B_a, \Delta.$$

Abbreviate $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, $C = q - F - \alpha$, and $R = k - 1 - hE$. These are substitutions, not new unknowns. The following 23 equations are quadratic after expansion:

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L+1), & X &= wq,\\
a &= E + sN, & E &= X(a - E), & c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1, & &\\
D &= X + ac + \gamma H, & \Delta &= a^2 + H, & C_2 &= c^2,\\
D^2 &= \Delta C_2 + 1, & G &= (q-1)(q-F) + \alpha, & &\\
R &= LG + (M_C + qM_F^{\mathrm{src}})J, & (K+w)C &= F + z(q-1), & &\\
F_2 &= f^2, & V &= of - c, & V &= jc - R,\\
g &= y^2 - V^2, & I &= iC_2, & B_a &= I^2,\\
B_a &= \Delta(F_2 - 1), & y^2 &= B_a g + 1. & &
\end{aligned} \tag{28}$$

The expanded residuals are supplied in `quadratic/quartic32.json`. Taking the sum of their squares gives an exact quartic: only the first norm residual contains τ, and its square contributes τ^4 with coefficient 1.

Proposition 8.1. *The positive solvability of* (28) *is equivalent to that of* (5).

Proof. The first equations give $Y = a - E = sq^3$, $E = XY$, $c = kY + \eta$, and $P = EkY + k$. Thus its first norm is exactly $\tau^2 - XY^2(XY^2 + 1)k^2 = 1$. The main, ordinary strong, packing, and transport equations are the same. The last equations give $I^2V^2 - (I^2 - 1)y^2 = 1$ with $V = of - c = jc - R$.

The ordinary rank lemma gives $c^2 \mid f^2 - 1$. Set $T = (o + Rf)/c$. This is integral because

$$o + Rf = f(of + R) - o(f^2 - 1), \qquad of + R = (j+1)c.$$

It is positive, and $c(Tf - 1) - Rf^2 = of - c = V$, recovering the fifteen-coordinate system.

Conversely, use a canonical positive solution, or apply the positive auxiliary restoration argument of package 02. The expressions

$$o = cT - Rf, \qquad j = Tf - 1 - R(f^2 - 1)/c$$

are positive integers. For clarity, positivity follows from $U > Rf^2 + c$, $V > -U$, the auxiliary norm's gap $|V| > 1 \Rightarrow |V| \geq 2U - 1$, and the exclusion of $V = 0, \pm 1$ by $f > 2c$ and $V \equiv -c \pmod f$. Thus $V > 1$. Since $(U-1)(y^2 - V^2) = V^2 - 1$, the supplied $g = y^2 - V^2$ is positive. All other introduced registers have positive values by their displayed definitions. □

The ordinary strong norm removes the old intermediate Δc^2 from package 08's lift. Restoring o, j instead of $T, Tf-1$ then makes R occur only linearly, allowing its elimination by $R = k-1-hE$. These two changes account for the improvement from 34 variables/25 quadratics to 32 variables/23 quadratics.

9 The local language and blank-halting equivalence

The fixed machine is `candidate-k215/k215.tm`, whose SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

It has 215 working states, two tape symbols, a blank two-sided tape, start state 0, and a separate halt 215. The first transition is $(0,0) \mapsto (1,+1,93)$; the unique entering-halt transition is $(68,1) \mapsto (1,+1,215)$. For tableau padding only, append $(215,b) \mapsto (b,0,215)$.

The 439 cell symbols are ordinary symbols 0 and 1; headed symbols $2(q_0+1)+b$ for $0 \le q_0 \le 215$, $b \in \{0,1\}$; initialization symbols L, Q, R with codes 434,435,436; vertical boundary 437; and horizontal boundary 438. Initialization payloads are blank, with the start head at Q.

The package-06 center predicate uses only the prior, left, center, right, and following cells. It propagates the vertical-boundary bit in time; forbids adjacent vertical boundaries and consecutive horizontal boundary rows; makes horizontal boundaries uniform inside a strip; and makes the initial row exactly L^+QR^+. Away from boundaries it applies the literal three-neighbor machine update, forbidding multiple head arrivals and escaping heads. The last interior row may contain a head only in the halt state. The complete predicate is preserved unchanged in `compiler/blank_helical.py`; its boundary clauses and exact machine table are part of the finite coefficient definition.

9.1 Why five positions suffice

A cross is (u,l,c,r,d), in prior/left/center/right/following order. It is locally valid if the old nine-cell predicate accepts it with ordinary blank corners. This is equivalent to every choice of corners, because the predicate never reads them. The old selector count is therefore exactly 439^4 times the cross count.

Enumerate the following two crosses first:

$$S = (438,434,435,436,1), \qquad E_* = (438,0,437,438,437).$$

Then enumerate all other valid crosses lexicographically. S is the initialized start cross; its following center is the ordinary symbol 1 written by the first transition. E_* is locally valid at its vertical-boundary center, but cannot occur globally: the prior horizontal-boundary cell has vertical bit 0, whereas its following center has vertical bit 1. Its exclusion removes no genuine tableau.

The exact number of valid crosses is

$$\kappa = 38\,256\,607\,897. \tag{29}$$

The inherited thirteen-category census counts these crosses directly; the new audit replays each category against the exact predicate.

Let the right neighbor of cyclic cell i be $i-1$, and its following-time neighbor be $i-h$. Four equality families suffice:

$$r_i = c_{i-1}, \qquad c_i = l_{i-1}, \qquad d_i = c_{i-h}, \qquad c_i = u_{i-h}.$$

Applying them at neighboring indices gives

$$W_i = (c_{i+h}, c_{i+1}, c_i, c_{i-1}, c_{i-h}). \tag{30}$$

This holds even when cyclic indices coincide. Pullback along $(x,t) \mapsto -x-ht \pmod N$ therefore reconstructs a periodic plane satisfying the exact center predicate.

9.2 A marked periodic plane contains a finite halting run

At a Start, scan its initialization row to the left and right. The L-predecessor and R-successor rules cannot close a cycle through Q, so both scans meet vertical boundaries. Between them lies

a finite strip with one start head and all blank payloads. Vertical-bit propagation preserves this strip in time. The horizontal boundary bit is uniform across it. The time shift is a permutation of the finite cyclic index set, so the initial boundary row recurs. Before its first return, every row is the exact deterministic machine update; the head stays unique and cannot cross a boundary. The row before the return permits a head only in state 215. Hence K215 halts on blank tape.

Conversely, any finite blank halting run has rectangular presentations with independently padded space and time margins, followed by halt padding. The standard helical presentation gives a cyclic word with the four overlaps, one Start at the chosen origin, and no forbidden E_*. The detailed padding and actual-index tuning argument is the package-06 blank-helical proof, whose hypotheses depend only on the center predicate and these overlap identities. Removing corners does not alter those hypotheses.

10 Four exact, smaller coefficient recipes

10.1 A closed finite definition

The following construction fixes all four integers in (2). There is no free program index, input, or oracle. Its finite local predicate is the literal K215 predicate of Section 9; the complete transition table and predicate are included in the archive. The coefficient definitions terminate after bounded finite enumeration, although their binary expansions are infeasibly large.

Set $a_0 = 439$ and $\kappa = 38\,256\,607\,897$. Let $W_0 = S$, $W_1 = E_*$, with

$$S = (438, 434, 435, 436, 1), \qquad E_* = (438, 0, 437, 438, 437).$$

List the remaining valid crosses in lexicographic order on $\{0, \dots, 438\}^5$, excluding these first two. Coordinates are always prior, left, center, right, following. The count κ is the exact count before the old compiler's free-corner factor 439^4. The source census and the independent replay are included; no halting test is performed in this enumeration.

Let ℓ be the least integer at least two satisfying $2^\ell > \kappa + 1$, and put $A_0 = 2^\ell$. Define

$$\begin{aligned}
p_0 &= \max\left\{0, \left\lceil \frac{\kappa - 2\ell - 9a_0 - 4}{2} \right\rceil \right\}, & j_{\max} &= 5a_0 + 4 + p_0, \\
\mu &= A_0 - 2 + \sum_{j=1}^{j_{\max}} A_0^j, & m_0 &= 2\ell + 14a_0 + 8 + 2p_0, \\
n_0 &= m_0 + 1, & n_{\mathrm{dum}} &= n_0 - \kappa - 5a_0 - 4, \\
e_* &= \kappa + 12a_0, & M_0 &= \kappa + 15a_0 + n_{\mathrm{dum}}.
\end{aligned} \tag{31}$$

In this instance $n_{\mathrm{dum}} = 1$. The definitions retain the original geometric separation between selectors, copies, dummies, and anchors.

Use the ordered shape

$$\Sigma = ((0,1), (1,0), (1,1), (1,2), (2,1)).$$

For $(r, c) = \Sigma_z$, $0 \le z \le 4$, and $0 \le u < a_0$, set

$$e(r, c, u) = \kappa + 3a_0 + (2 - r)3a_0 + (2 - c)a_0 + u, \qquad j(z, u) = 1 + za_0 + u. \tag{32}$$

The allowed radix positions form the disjoint set

$$\begin{aligned}
\mathcal{E} = \{0, \dots, \kappa - 1\} \;\cup\; \{e(r, c, u) : (r, c) \in \Sigma,\; 0 \le u < a_0\} \\
\cup \{e_*, \dots, e_* + n_{\mathrm{dum}} - 1\} \;\cup \{M_0, 3M_0, 9M_0, 27M_0\}.
\end{aligned} \tag{33}$$

It has cardinality n_0. The five copy blocks, in the order Σ, start at $\kappa+10a_0, \kappa+8a_0, \kappa+7a_0, \kappa+6a_0, \kappa+4a_0$; each has length a_0. All nonanchor positions lie below M_0.

Define clause coefficients c_e, for $e \in \mathcal{E}$, by

$$\begin{aligned} c_i &= 1 + \sum_{z=0}^{4} A_0^{1+za_0+W_i[z]} + \sum_{t=1}^{4} A_0^{5a_0+t} && (0 \le i < \kappa), \\ c_{e(r,c,u)} &= A_0^{j(z,u)} && ((r,c) = \Sigma_z), \\ c_{3^{t-1}M_0} &= A_0^{5a_0+t} && (1 \le t \le 4), \\ c_e &= 0 && \text{at dummy positions.} \end{aligned} \tag{34}$$

The leading 1 counts selected crosses; the other digits pair each selected cross with its five copies and four anchors.

10.2 Band separation, radix, and the four ports

Put

$$\begin{aligned} E_{\max} &= 27M_0, & H_C &= 51M_0 + 3a_0 + 1, \\ T_1 &= H_C + 2E_{\max} + a_0 + 1, & T_2 &= T_1 + 4E_{\max} + 2. \end{aligned} \tag{35}$$

Let L_0 be the least power of five at least $T_2 + 2E_{\max} + 2$. For $v > 0$, let $\mathrm{bits}(v) = \lfloor \log_2 v \rfloor + 1$, and set

$$\begin{aligned} U_0 &= 4(n_0+2)\big(28(\kappa+5a_0+4)+6\big)+8, \\ b_{\min} &= \max\big\{\ell(j_{\max}+1)+3,\ \ell(5a_0+4)+\mathrm{bits}(U_0)+1,\ 4\big\}. \end{aligned} \tag{36}$$

Let b be the least power of five at least $b_{\min}$, and put

$$\rho = 2^b, \qquad d = bL_0, \qquad B = \rho^{L_0} = 2^d.$$

All exponents used below are nonnegative.

The transport coefficients are

$$\begin{aligned} D_C &= \rho^{3a_0} + \rho^{H_C+a_0} + \rho^{8M_0} + \rho^{24M_0} + \sum_{e\in\mathcal{E}} c_e\big(\rho^{T_1-e} + \rho^{T_2-e}\big), \\ D_R &= \rho^{H_C}. \end{aligned} \tag{37}$$

Define the unshifted field mask by

$$\begin{aligned} M_F^0 = \mu(\rho^{T_1} + \rho^{T_2}) &+ \sum_{u=0}^{a_0-1} \big(\rho^{e(0,1,u)} + \rho^{e(1,1,u)}\big) \\ &+ \sum_{u=0}^{a_0-1} \big(\rho^{H_C+e(1,0,u)} + \rho^{H_C+e(1,1,u)}\big) + \rho^{9M_0} + \rho^{27M_0}. \end{aligned} \tag{38}$$

Finally set

$$M_C = B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\ne 1}} \rho^e - 2\rho^{e_*}, \qquad M_F = M_F^0 + 4. \tag{39}$$

The four fixed polynomial coefficients are exactly

$$\boxed{\beta = B-1, \qquad K = D_C + BD_R, \qquad M_C, \qquad M_F^{\mathrm{src}} = M_F + B - 1.} \tag{40}$$

In particular, the source port M_F^{src} is not the native mask M_F. Every arithmetic presentation in this article uses this same quartet.

The concrete scalar geometry is

Quantity	Value	Quantity	Value
ℓ	36	n_{dum}	1
p_0	19 128 301 935	$j_{\max}$	19 128 304 134
m_0	38 256 610 096	M_0	38 256 614 483
e_*	38 256 613 165	$E_{\max}$	1 032 928 591 041
H_C	1 951 087 339 951	T_1	4 016 944 522 473
T_2	8 148 658 886 639	$b_{\min}$	688 618 948 863

Thus

$$b = 5^{17}, \qquad L_0 = 5^{19}, \qquad d = 5^{36} = 14\,551\,915\,228\,366\,851\,806\,640\,625. \tag{41}$$

The coefficient β has exactly d binary digits. The corresponding package-06 value was 5^{67}; the reduction factor is 5^{31}. This improves coefficient height but does not make expansion feasible. The new recipe also removes four copy coordinates and eight families of overlap tests.

10.3 Populations, carry bounds, and exact convolution tests

Lemma 10.1. *The coefficients above satisfy*

$$0 < M_C, M_F < B - 1, \qquad M_C \equiv 2 \pmod 4, \qquad M_F \equiv 4 \pmod 8,$$
$$\mathrm{pc}(M_C) + \mathrm{pc}(M_F) = d.$$

They implement the four cross overlaps of Section 9 with the same carry-free native interface as the package-06 compiler.

Proof. The powers of A_0 occupy disjoint ℓ-bit blocks, so

$$\mathrm{pc}(\mu) = \ell + 5a_0 + 3 + p_0, \qquad m_0 = 2\,\mathrm{pc}(\mu) + 4a_0 + 2.$$

The two clause sites in M_F^0 contribute $2\,\mathrm{pc}(\mu)$; the four copy-test families contribute $4a_0$; and the anchor sites contribute two. Hence $\mathrm{pc}(M_F^0) = m_0$ and $\mathrm{pc}(M_F) = m_0 + 1$. The added low bit 4 is disjoint from all other support.

In M_C, exactly n_0 distinct bits are removed from $B - 1$: one low bit at every native position except selector 1, plus one upper bit at the first dummy. No borrow occurs. Thus

$$\mathrm{pc}(M_C) = d - m_0 - 1, \qquad \mathrm{pc}(M_C) + \mathrm{pc}(M_F) = d.$$

Selector zero is permitted, selector one is forbidden, ordinary native digits are Boolean, and the first dummy digit may be $0, 1, 2, 3$. The lowest mask bits give the displayed congruences; unused high positions give the strict range bounds.

The clause coefficients obey the conservative estimate

$$\sum_{e \in \mathcal{E}} c_e \le 14(\kappa + 5a_0 + 4)A_0^{5a_0+4}, \qquad \mu < A_0^{j_{\max}+1}.$$

Consequently the radix definition implies

$$\rho \ge \max\left\{4(n_0+2)\left(2\sum_e c_e + 6\right) + 8,\ 2\mu + 4,\ 16\right\}. \tag{42}$$

This bounds every unshifted native field coefficient by $\rho/4 - 2$, including the permitted upper dummy bit. Moreover $T_2 + E_{\max} < L_0$, so no unshifted field term spills across a cell boundary.

At each clause site T_j, convolution by D_C reads exactly $\sum_e c_e x_e$ from the current cell. The other clause band cannot reach it because $T_2 - T_1 > 2E_{\max}$. The short-shift terms lie strictly below the clause bands.

At $e(0,1,u)$ and $e(1,1,u)$, the short shift $3a_0$ reads, respectively, the current center and following copies. The identity contribution from the temporal neighbor reads its prior and center copies. These are the two temporal overlaps.

At $H_C + e(1,0,u)$ and $H_C + e(1,1,u)$, the shift $H_C + a_0$ reads the current center and right copies, whereas D_R reads the spatial neighbor's left and center copies. These are the two spatial overlaps.

At $9M_0$ and $27M_0$, the shifts $8M_0$ and $24M_0$ read current anchors M_0 and $3M_0$; the temporal neighbor contributes its anchors $9M_0$ and $27M_0$. There are no unintended terms. Low support is below M_0; the four anchors are separated by powers of three; high clause bands lie above the test sites; and $H_C = 51M_0 + 3a_0 + 1$ separates horizontal tests from the possible anchor sum $24M_0 + 27M_0 = 51M_0$. These support statements prove the exact convolution roles for every word, not merely for the toy audits. □

10.4 Why arbitrary bit rotations still synchronize

We give the synchronization argument because it is the principal compiler obligation beyond the population identity. After arithmetic soundness recovers $X = 2^R$, write its binary rotation exponent as $bt + \epsilon$, $0 \le \epsilon < b$. The possible rotated radix support is contained in a translate, modulo L_0, of

$$\mathcal{E} \cup \{e_* + 1\} \subset [0, E_{\max} + 1].$$

The extra position records the upper dummy bit spilling into the next radix position. Both cyclic distances between T_1 and T_2 exceed $E_{\max} + 1$, so this rotated support cannot contaminate both clause sites. One clause site is clean.

For $\epsilon \le b - 2$, a rotated digit is at most $3\rho/4$. For $\epsilon = b - 1$, it is at most $\rho/2 + 1$. Together with (42), neither case creates a carry. Thus the mask tests apply to the actual field. More explicitly, the transport equation and $X = wq \equiv w \pmod{q-1}$ identify F modulo $q - 1$ with the native field $\widehat{F}$. The carry bounds give $0 < \widehat{F} < q - 1$, while arithmetic soundness gives $0 < F < q$; therefore $F = \widehat{F}$.

At the clean clause site, the low base-A_0 digit counts selected crosses. This count is at most $\kappa < A_0 - 1$, and its AND with $A_0 - 2$ vanishes only for counts zero and one. Every higher clause digit is a sum of the relevant selected indicator and one copy or anchor bit. These sums are below A_0, so no base-A_0 carry occurs. Their masked low bits force evenness. Hence an occupied cell has exactly one selector, all five matching copies, and all four anchors; an unoccupied cell has none of these copies or anchors. Dummy bits have coefficient zero and remain independent.

The marked origin is occupied. Its two anchor tests require two distinct rotated digits to be odd. If $1 \le \epsilon \le b - 2$, all rotated digits are even. If $\epsilon = b - 1$, only the single upper-dummy spill position can be odd. Thus $\epsilon = 0$.

All nonanchor native positions lie below M_0, and the only pair of native positions separated by $18M_0$ is $(9M_0, 27M_0)$. Since $L_0 > 54M_0$, reduction modulo L_0 cannot supply another such pair through wraparound. The two anchor tests therefore force $t = 0 \pmod{L_0}$: the surviving rotation is by a whole number of cells.

The copy tests now propagate occupancy to both spatial neighbors. Indeed, a selected cross has one-hot copies, and its tested center or side copy cannot match an empty neighbor. The spatial offset 1 generates the whole cyclic word, so Start forces every cell occupied. The four decoded overlaps then yield (30). In particular a Start-containing solution cannot have zero temporal stride, since its prior cell and center are different. No uniqueness of Start, or independent End marker, is required for soundness.

10.5 Completeness at the actual packed index

The preceding support argument also preserves the canonical completeness construction of package 06 [4]. We record the index-control step explicitly, rather than assuming that a formal congruent exponent can replace the actual packed value.

Choose a padded halting tableau with spatial width h_0 and temporal height H_0, both sufficiently large powers of five, such that $H_0 \geq 25$ and $N = h_0 H_0 > 25d$. The helical quotient has one Start at the origin. Set $q = B^N$. Initially set every dummy bit to zero and let C_0, F_0, R_0 be the actual native word, actual field, and actual packed index. Switching on the low dummy at an eligible cell v changes

$$\begin{aligned} \delta C &= \rho^{e_*} B^v, \\ \delta F &= (D_C + BD_R + B^{h_0})\rho^{e_*} B^v, \\ \delta R &= -\Gamma B^v, \qquad \Gamma = \rho^{e_*}(q^2 - 1)[1 + q(D_C + BD_R + B^{h_0})]. \end{aligned} \tag{43}$$

Use distinct positions

$$\{4j : 0 \leq j < N/5\} \cup \{1\}.$$

Neither spatial nor temporal movement wraps these individual contributions, because $h_0 = N/H_0 \leq N/25$.

The two clause-band terms in D_C cancel modulo five:

$$T_2 - T_1 \equiv 2 \pmod 4, \qquad \rho \equiv 2 \pmod 5.$$

Since $a_0 \equiv 3 \pmod 4$,

$$2D_C - D_R \equiv 2(2 + 3\rho^{H_C} + 1 + 1) - \rho^{H_C} \equiv 3 \pmod 5. \tag{44}$$

Also $q, B, B^{h_0} \equiv 2 \pmod 5$, so the bracket in (43) is $2D_C - D_R$ modulo five. Thus Γ is invertible modulo the power of five dN. No conditional correction monomial is needed.

Every residue modulo dN is a Boolean subset sum of the displayed weights B^v. To see this, B^4 has order $N/5$ modulo dN, by

$$\nu_5(2^{4dj} - 1) = \nu_5(d) + 1 + \nu_5(j).$$

Its powers enumerate exactly the residues $1 + 5dt$, with t modulo $N/5$. For a desired residue, optionally subtract the distinct weight B so that the remaining residue is nonzero modulo five. Let k_0 be its least positive residue modulo $5d$. Then $1 \leq k_0 < 5d < N/5$ and k_0 is a unit modulo $N/5$. A cyclic interval of k_0 distinct residues t has sum $k_0 t_0 + k_0(k_0 - 1)/2$; varying t_0 gives every sum modulo $N/5$. Choosing the corresponding powers of B^4 proves the subset assertion.

Apply it to $(R_0 - dh_0)\Gamma^{-1} \pmod{dN}$. After this Boolean adjustment the *actual* packed index obeys

$$R \equiv dh_0 \pmod{dN}. \tag{45}$$

The dummy is absent from every clause and copy test; its upper bit remains zero. All masks survive the adjustment. The canonical digit bounds also give

$$2C + F \leq (\rho/4 + 1)\frac{q - 1}{\rho - 1} < q/3,$$

so the outer slack $q - F - C$ stays strictly positive.

The shifted field mask deserves a separate check. Its unit radix digit comes only from the temporal neighbor's Start selector, since D_C, D_R have positive minimum exponent. Canonical Start occurs only at zero and $0 < h_0 < N$. That digit is therefore zero at the origin and zero

or one elsewhere. The origin's two low mask bits and the other cells' bit 2 vanish as required. Dummy adjustment occurs at positive radix exponent and changes none of these tests.

Consequently the inverse-population identity and the native masks give, at this same final R,

$$3q + 1 < R < q^4, \qquad R \equiv 3 \pmod 4, \qquad \mathrm{pc}(R) = 3dN + 2.$$

The positive half-binomial converse [9, 4] now supplies fresh Pell witnesses at the actual index. It gives $X = 2^R$, $q^3 \mid Y$, and hence $w = X/q \in \mathbb{Z}_{>0}$. Equation (45) yields $w \equiv B^{h_0} \pmod{q-1}$. The native transport quotient

$$z = \frac{(K + w)C - F}{q - 1}$$

is integral and positive, since $w > q > F$ and $C \geq 1$. This is the shifted transport convention of (5). The ordinary strong witness is obtained from the canonical normalized one by multiplication by Δ. Thus the smaller coefficient quartet preserves completeness.

10.6 Scope and reproducible evidence

The inherited dependencies are the marked helical-tableau argument, the native carry and mask interpretation, and the half-binomial Pell converse [4, 10, 9]. The archive also preserves the native 76-operation compiler proof, the half-binomial mask correction, and the five-adic subset-control note, with exact source identities. The present change is the five-coordinate language and its revised finite layout; it does not silently specialize a theorem whose input-marker hypotheses remain unmet.

The generator `compiler/cross_constants.py` emits the closed recipe, all scalar geometry, and its audits. With `-tm`, it checks the exact source-table hash, all thirteen census categories, 32 helical control presentations, and 1,144 projected crosses. Its other checks include 26,106 cyclic lift checks, 100,384 exhaustive small cross-word candidates, and 16,224 sparse convolution basis tests. An independent audit rederives the large-instance geometry and support isolation by interval arithmetic without importing the generator.

These finite checks supplement the parametric proofs above. They do not enumerate the actual selector alphabet, expand the coefficients, exhibit a halting K215 run, or kernel-check the Diophantine equivalence. All definitions and proofs are ordinary finite mathematics formalizable in ZFC; supplying such an argument is distinct from supplying an encoded ZFC derivation or a new Lean/Rocq theorem.

11 Unrestricted integer equations and exact degrees

The fifteen-variable arithmetic proof required positivity of exactly the nine coordinates in (8). Replace each of these by its own disjoint copy of

$$\mathrm{Pos}(u, v, t) = 1 + u^2 + v^2 + t(t + 1). \tag{46}$$

This is always positive on integers and represents every positive integer. For surjectivity, apply the three-square theorem to $4(n-1)+1$, which is not of the excluded form $4^a(8b+7)$. Exactly one of its three squares is odd; writing it as $(2t+1)^2$ and the others as $(2u)^2, (2v)^2$ gives (46). The remaining six coordinates stay unrestricted. There are therefore $9 \cdot 3 + 6 = 33$ integer unknowns.

The quadratic system likewise requires positivity of only $J, F, \alpha, z, i, s, w, \eta, \zeta$. Its other 23 coordinates may be arbitrary integers: the scale equations force the required positive scales, the signed auxiliary lemma handles o, j, V, f, y, and the main-root argument handles D, γ. Thus (46) gives $9 \cdot 3 + 23 = 50$ unrestricted unknowns. Its residual degrees are at most 4 and its sum of squares has exact degree 8. The first residual has a nonzero quartic leading form, and a sum of real squares of leading forms cannot cancel identically.

Integer presentation	Unknowns	Equations	Degree	Operations
Gap root norm	32	1	434	138
Root outer norm	32	1	680	136
Lifted $\mathcal{S}_{60}$	33	6	50 maximum	123
Lifted gap system	33	6	44 maximum	124
Lifted P_{69}	33	1	190	132
Lifted gap product	33	1	179	133
Lifted S_{77}	33	1	100	140
Lifted gap sum of squares	33	1	88	141
Lifted $\mathcal{Q}_{32}$	50	23	4 maximum	not optimized
Its sum of squares	50	1	8	not optimized

11.1 Degree certificates for the short circuits

Give each positive witness degree one and fixed numerals degree zero. Let $Q = \beta J$, $k_0 = \eta + \zeta$, and $C_1 = Q - F - \alpha$. The top parts of X, Y, a, c are $wQ, sQ^3, wsQ^4, k_0 sQ^3$. The six factors of P_{69} have exact degrees

$$22, \quad 18, \quad 28, \quad 7, \quad 2, \quad 22.$$

The main norm has its leading nominal terms cancel; expanding it gives

$$D^2 - \Delta c^2 = 2ac(X + \gamma H) + (X + \gamma H)^2 - Hc^2,$$

whose leader is $8\gamma(wsQ^4)^2(k_0 sQ^3)$. Multiplying the nonzero factor leaders gives

$$8Q^{55}h\gamma i^2 k_0^9 w^7 s^{16} T^2 f^4 (wC_1 - z_0 Q), \tag{47}$$

a nonzero degree-99 form for every admissible $\beta > 0$. For the sum of squares, the auxiliary residual uniquely has degree 28, so its squared leader proves exact degree 56.

Under the mixed integer conversion, the guarded variables have weight two and the six signed variables weight one. The factor degrees become $44, 35, 50, 13, 4, 44$, totaling 190; the maximum residual degree is 50. Substitution of the disjoint leading forms $u^2 + v^2 + t^2$ is injective: set two coordinates in every triple to zero to obtain the injective monomial map $x \mapsto u^2$. Thus these weighted degrees are attained, not merely upper bounds or random-specialization guesses.

12 ZFC calibration and verification status

At the pinned snapshot, the accepted machine endpoint is

```
KEndpoint215.blank_halts_iff_zfc_refutation:
BlankHalts machine215 ↔
BProv ZFCax_s [] Form.fBot.
```

The negated endpoint is `KEndpoint215.nonhalting_iff_zfc_consistency`. The verification dossier records Lean and Rocq builds for the same literal 430-row machine; their source and table identities are distinct from the new Diophantine work [11, 12].

The ordinary calibration argument turns this exact blank computation into a finite refutation predicate for the specified ZFC presentation. Standard effective proof presentations translate into one another by finite proof translations formalizable in ZFC. Every construction above uses finite combinatorics, integer arithmetic, induction, Pell sequences, and explicitly bounded coefficient enumeration. The arithmetic and compiler proofs, combined with the ordinary machine calibration, are therefore formalizable in ZFC and give

$$\mathrm{ZFC} \vdash [\exists \text{ a solution}] \leftrightarrow [\exists p \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)].$$

Negation gives (1). This does not infer ZFC-provability from Lean's soundness; it uses the ordinary mathematical arguments that the machine formalization mirrors.

No new kernel-checked proof of the native compiler, Pell bridge, or present equivalence is included. The full Lean/Rocq builds were not rerun. The computational checks establish exactly their advertised finite properties: literal arithmetic identities, live gate counts, degree witnesses, sparse convolution maps, small cyclic reconstruction, the exact category census, and the source table hash. They do not search for an accepting K215 run, materialize the giant coefficients, or decide consistency.

No consistency assumption is needed to prove the biconditional. If ZFC is consistent, its second incompleteness theorem prevents ZFC from proving the resulting unsolvability statement. Mere consistency alone should not be silently strengthened to a claim of two-sided independence; an additional appropriate soundness hypothesis is needed to exclude a proof of the false existential statement.

13 Reproduction, source attribution, and further work

The archive is self-contained for its finite checks and uses Python's standard library. Run `python3 reproduce.py` from its root to regenerate the arithmetic packets, quadratic residuals, coefficient specification, and verification receipts. Run `make report` to compile this article with pdfLaTeX. The main source and four included TeX sections must stay together. A file manifest records SHA-256 values. The fresh seven-stage run authenticates 24 exact upstream files and emits 20 arithmetic presentations, including retained intermediate tradeoffs. Its independent sparse-polynomial audit proves the six auxiliary-gap identities by exact symbolic expansion, without sampling. The independent interval audit covers every actual native position, not just the small control instances.

The proof dependencies retain their original filenames and attribution. This contribution uses the project's MIT No Attribution terms and preserves the inherited K215 notices, including Andrew James Wade's compiler ancestry. The principal mathematical credit is to the eight source packages and the underlying native compiler development; the present contribution is a checked synthesis with the additional cross-language, transport, lift, and signed-domain simplifications described explicitly above.

13.1 Further research questions

1. **Formalize one interface end to end.** A focused next theorem is the signed auxiliary lemma with its exact hypotheses, followed by the ordinary strong-rank lemma. The five-cell compiler can then target the accepted K215 endpoint without importing any consistency premise.
2. **Quotient the prior-cell information.** The center predicate reads the prior cell only through its horizontal-boundary bit. A heterogeneous representation with four full-cell fields and two boundary-bit copies would have only 87 923 161 local patterns. The corresponding sparse layout and carry/synchronization proof have not been completed here. This is a concrete next coefficient target, not part of the claimed coefficient recipe.
3. **Directly compile the numerical calculus.** A local language for the calibrated C222 proof search could avoid the 215-state machine alphabet entirely. Such a change needs its own exact proof-predicate calibration and cannot be justified only by an equiconsistency slogan.
4. **Optimize the fourteen-variable norm.** Its root resultant has high degree. The ordinary strong norm offers smaller degrees but does not permit the same immediate integral-square

argument; any attempt to use a rational discriminant must recover integrality.

5. **Reduce the remaining guarded coordinates.** The proof already releases six signs in the fifteen-variable form. Releasing i without excluding zero is invalid: it collapses the strong and auxiliary conditions. A valid further reduction must address that degenerate branch explicitly.

6. **Measure coefficient descriptions as well as gate counts.** The five-cell change substantially reduces actual height while leaving the gate count unchanged. Compact symbolic sums, effective smaller alphabets, and bounds that charge coefficient bit length should be reported alongside arithmetic operation totals.

7. **Internalize the equivalence in weaker arithmetic.** The present claim is ZFC-provability by formalizable ordinary mathematics. A claim in PA or a weaker system requires a separate arithmetization and accounting for the induction used in the compiler and Pell lemmas.

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